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M-13 bacteriophage based structural color sensor for detecting antibiotics

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ABSTRACT

Color sensor systems that exploit the advantages of M-13 bacteriophage have been shown to be potentially useful for detection of hazardous materials. The properties of M-13 bacteriophage can be systemically modified to impart target-specific selectivity and sensitivity using the phage display technique. Here, we describe a structural color-based sensor that utilizes genetically engineered M-13 bacteriophage to discriminate different types of antibiotics. An M-13 bacteriophage based structural color matrix was fabricated using a simple pulling technique by self-assembly of M-13 bacteriophage. When exposed to organic solvent, M-13 bacteriophage bundles promptly swell and promote distinct structural color change. Color sensors composed of M-13 bacteriophage genetically engineered to possess WHW peptide motifs clearly discriminated three different types of antibiotics, which was based on the color analysis of sensor using principal component analysis. Our sensing approach based on M-13 bacteriophage could be a promising sensor technique such as an environmental monitoring system.

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1. Introduction

Recently, public health and environmental pollution have been attracted and get much attention due to concerns about pathogenic agents, hazardous chemical leakage. Furthermore, abusing antibiotics is detrimental to public health and creates environmental pollution [1–4]. Accordingly, the development of detection tools that can selectively and sensitively detect low concentration of chemicals (ppm level) has drawn much research attention. As a result, colorimetric sensors [5,6] utilizing various materials and physical phenomenon have been developed [7–12], but sensitivity and selectivity remain issues of concern because, in most of cases, satisfying both demands simultaneously is highly challenging. Here, we describe a highly sensitive and selective color sensor system based on M-13 bacteriophage for detection of the antibiotics.

M-13 bacteriophage is a filamentous bacteriophage that is mainly composed of 2700 copies of helically arranged pVIII major coat protein. Typical M-13 bacteriophage is highly regular with a length of ~880 nm and a diameter of ~6.6 nm. Since foremost parts of the sequences of major and minor coat proteins are well known, M-13 bacteriophage can be easily modified genetically to confer different properties, such as, target-specific binding affinity. In addition, to its regular shape and genetic features, self-assembly is an important characteristic of M-13 bacteriophage. Using a simple pulling technique and self-assembly, M-13 bacteriophage can be assembled to form a structural color film due to its liquid crystalline-like behavior [13–19]. The performances of M-13 bacteriophage have been successfully demonstrated by various sensor platforms, such as, fluorescence [18], surface plasmon resonance (SPR) [20] and color [13,14] sensors. In particular, M-13

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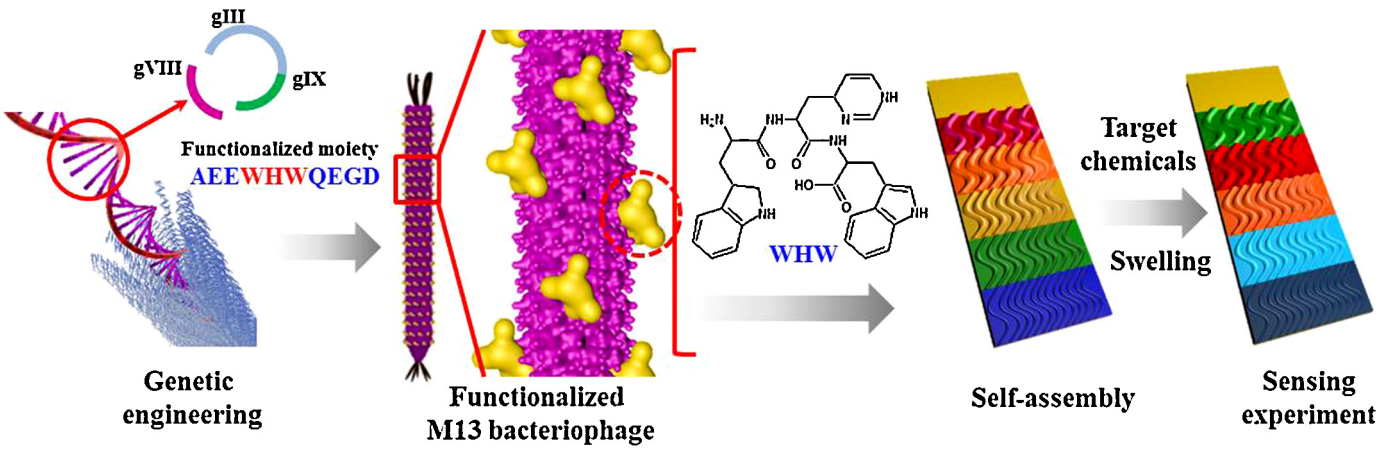

Fig. 1. Schematic of the fabrication of the M-13 bacteriophage-based color sensor and measurement of target chemicals. Color sensors were fabricated through a simple pulling technique using genetically engineered M-13 bacteriophage. Applying target chemicals to color sensor causes structural color tuning due to the structural modification resulting from the adjusted coherent scattering following bundle distance change.

bacteriophage-based color sensors are well regarded due to their ease of handling and rapid responses.

In this study, genetically engineered M-13 bacteriophage was utilized to fabricate a color sensor. Evolutionary screening revealed a functionalized M-13 bacteriophage with the AEEWHWQEGD (*Ala(A)-Glu(E)-Glu(E)-**Trp(W)-His(H)-Trp(W)**-Gln(Q)-Glu(E)-Gly(G)-Asp(D)*, WHW) peptide sequences in pVIII proteins (WHW bacteriophage) from the protein sequence of pVIII in wild type of M-13 bacteriophage (*Ala(D)-Glu(E)-Gly(G)-Asp(D)-Asp(D)*). This WHW bacteriophage displays about 2700 copies of Tryptophan (W)- Histidine (H)-Tryptophan(W) sequences at the end of pVIII proteins, which means dense and regular receptor for target materials. WHW bacteriophage can specifically bind to aromatic group-containing chemicals, like antibiotics, via π-π interactions with the aromatic rings of its Tryptophan and Histidine residues. A quasi-ordered M-13 bacteriophage structure was formed in a bundle assembly, and color of each bands were tuned by changing film extraction speeds from an aqueous bacteriophage solution. Amoxicillin and rifampicin were chosen as target antibiotics based on considerations of their environmental contamination potentials. The commercial antibiotics were paired with pure chemicals of its main components. Our data reveals that M-13 bacteriophage-based color film can be used as a possible chemical detecting sensor with promising reliability.

2. Material and methods

Fig. 1 provides a schematic of sensor fabrication. Genetically engineered M-13 bacteriophage was used to fabricate color sensors by self-assembly. Contacting target material with bacteriophage-based color sensor promotes the structural color change of bacteriophage bundle [13,14]. Real-time color changes were measured versus initial color, and then reproduced as color bands using RGB color components for visualization.

2.1. Fabrication of the M-13 bacteriophage-based color sensor

We fabricated self-assembled color band using a simple pulling technique [13,14]. A Au-coated Si wafer (525 μm thicks, 100 nm of Au over a titanium adhesion layer, Platypus Co.) was dipped into a bacteriophage solution (WHW M-13 bacteriophage) and pulled out. For preparing M-13 bacteriophage color film, the bacteriophage concentration in the solution was 5 mg/mL in tris-buffered saline (12.5 mM tris and 37.5 mM NaCl, pH 7.5). Color bands on the sensor were controlled using different gradual pulling speeds from 10 to 100 μm/min by a programmable syringe pump (KD Scientific Co., LEGATO270). Using this approach, three distinct colors were generated on film surfaces. Each color band was obtained through different pulling speeds. Atomic force microscope (AFM) images were obtained using NX10 AFM (Park systems, Suwon, Korea) and analyzed using XEP 3.0.4 (Park systems, Suwon, Korea) and XEI 1.8.2 (Park systems, Suwon, Korea). All images were taken in true non-contact mode. Fast Fourier Transform (FFT) analysis of AFM height profiles was done to study the periodicity of the bundle structure. Numerical computation of the Fourier transform was carried out using a 2D FFT algorithm in XEI 1.8.2.

2.2. Detection of organic solvents

Detection of samples by M-13 bacteriophage color sensor was performed in a home-built detection system [13]. All chemicals were purchased from Sigma-Aldrich and used without further purification. Solvent detection experiments were performed by injecting the required volumes of solvents (10, 30, 50, and 100 ppm) into a 500 ml home-built sealed chamber with heating block through an inlet tube at 30 °C. M-13 bacteriophage-based color sensor (WHW type) was placed on the top of chamber fitted with a CCD microscope camera (Celestron, LLC.) controlled by MATLAB (MathWorks, Inc.) to display real-time RGB data and for image capturing. The color sensor was wholly covered except small ventilation hole to eliminate external light interference. We fixed the gain of digital microscope to prevent unwanted compensation of brightness. The regions of interest at each color band were chosen. All data, including real-time color changes in RGB signals were acquired, and color images were captured usually every 10 s. Measurements were repeated at least 3 times, and mean and standard error were calculated. Color changes were visualized using Photoshop program (Adobe Systems Inc.) by calculating RGB color differences between initial and final images.

2.3. Detection of antibiotics

Cephalosporin class antibiotics containing 500 mg of cefadroxil was purchased from Boryung Co., Ltd. Penicillin class antibiotic containing 500 mg of amoxicillin was purchased from Chong Kun Dang Pharmaceutical Corp., and rifampin class antibiotic containing 600 mg of rifampicin was purchased from CJ HealthCare Corp. All of commercial antibiotics were used without purification. Cefadroxil,

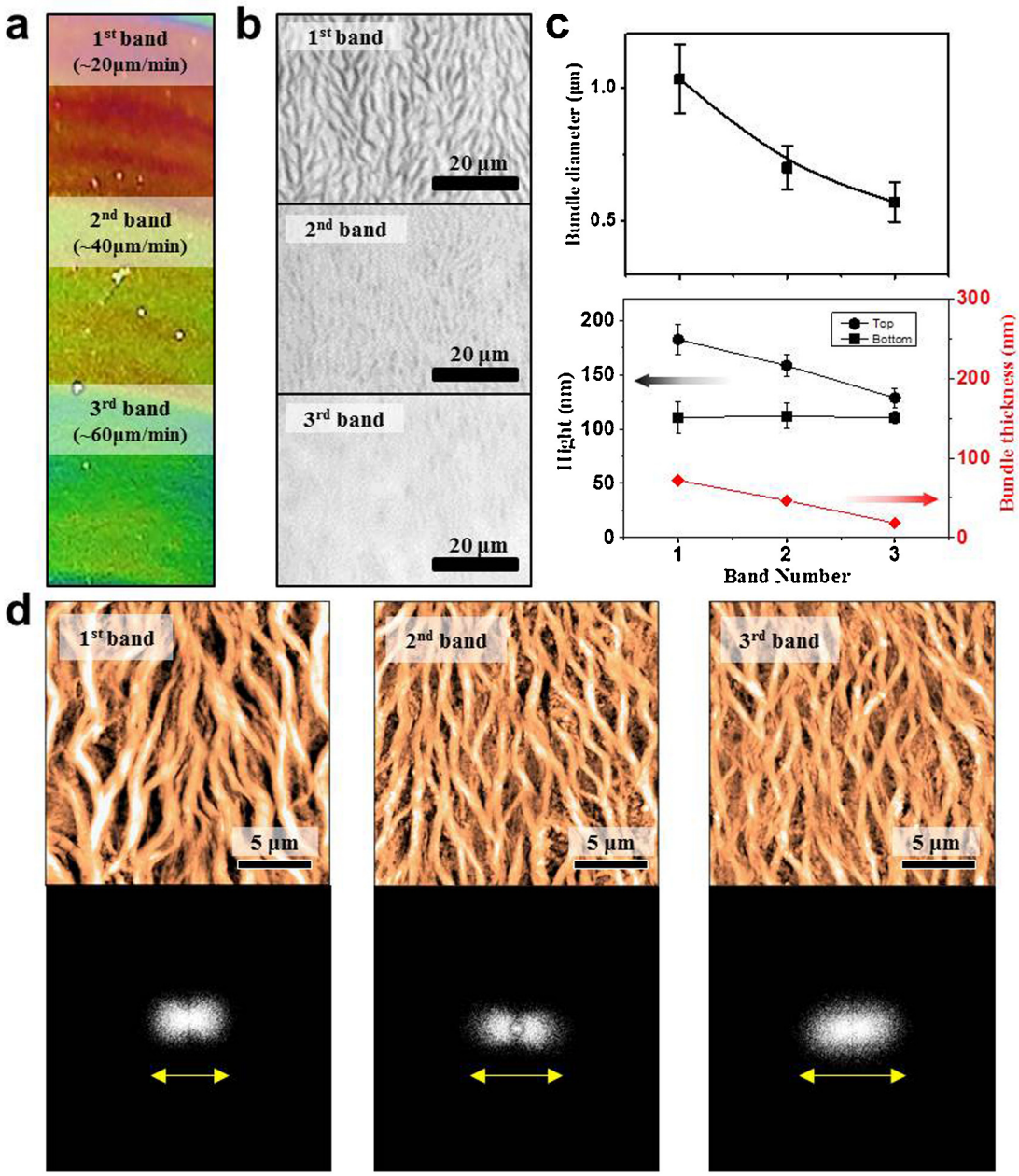

Fig. 2. Characterization of M-13 bacteriophage color sensor through atomic force microscopy (AFM) and fast Fourier transform (FFT) analysis. (a) Image of structural color of M-13 bacteriophage color sensor through CCD microscope camera. (b) Optical microscope image of bundles. (c) Modulation of M-13 bacteriophage bundle structures at different color bands. The effects of various pulling speed (20–60 μm) on bundle (left) height (top and bottom) and (right) thickness. (d) AFM images and two-dimensional FFT analysis of bacteriophage bundles at different color bands.

amoxicillin and rifampicin reagents were purchased from Sigma-Aldrich and used without further purification. WHW type M-13 bacteriophage was used for the experiment. Unlike the solvent experiment, measurements of solid and commercial antibiotics and pure components were accomplished using an excess amount of each sample except rifampicin. We used 500 mg each for all samples except rifampicin commercial antibiotics (416.7 mg) because of the different amount of ingredient. Commercial antibiotics were carefully ground to fine powders. The solid samples were placed in the sealed chamber for each experiment, and the chamber temperature was then increased in steps from 30 to 60 °C, 60 to 90 °C, and 90 to 120 °C at every 6 min to achieve elevated vapor concentrations. We assumed that the vapor concentration was proportionally related with the temperature of a tested sample. The real-time data were collected in the same manner as performed in the measurement of organic solvents.

3. Results and discussion

3.1. Characterization of M-13 bacteriophage color sensor

An array of hierarchical M-13 bacteriophage structure on solid substrate (Au-coated Si wafer) was fabricated through controlled pulling technique. By different extraction speeds of solid substrate from a solution of M-13 bacteriophage solution, we could fabricate distinct and tunable structural color nanostructure. Fig. 2a shows clear color band construction by this approach without any additional structural component such as polymer. Optical micro-

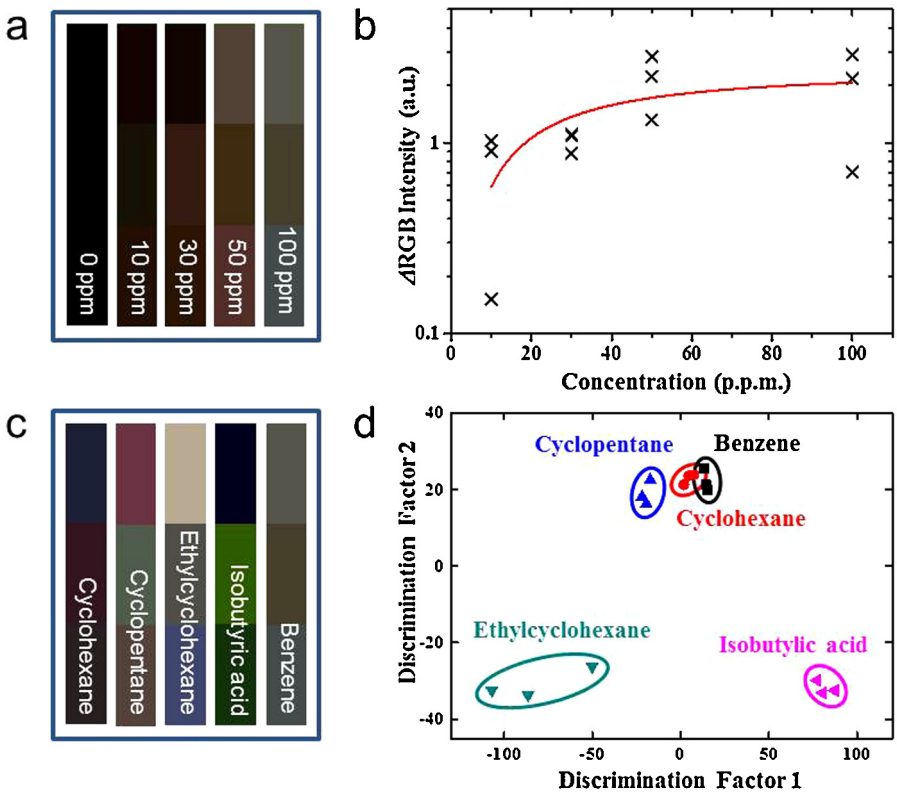

Fig. 3. Measurement of organic solvents using the M-13 bacteriophage-based color sensor. (a) Measurement of benzene at different concentrations in the gas phase from 10 to 100 ppm, (b) the most distinct increasing color signal of benzene detection at each band were fitted with Hill's equation to obtain a dissociation constant (K_d), (c) Various organic solvents were examined at 100 ppm, (d) 2D PCA score scatter plots (the first vs. second principal components) for the discrimination of different solvents.

scope image in Fig. 2b shows different bundle structures of each band by different extraction speeds. The self-assembled bacteriophage bundles are similar to the collagen fiber bundles in avian and mammalian skins that have structural color property [13,21,22]. This is one of the most distinct characters of bacteriophage-based bundle structure which differentiates from typical synthetic photonic crystal structure such as crystalline arrays or multilayer thin films [23,24]. AFM and FFT analysis show the contribution of the fiber bundle structure to observe structural color result from coherent scattering (Fig. 3c and d). The quasi-ordered bundle structure of M-13 bacteriophage exhibits angle-independent structural color by average bundle distance which is similar to quasi-ordered dermal protein arrays in animals and insects [25–27]. The M13-bacteriophage based color sensors were stable for months under ambient conditions.

3.2. Detection of organic solvents

Prior to detection of antibiotics, we measured common organic solvents using the M-13 color sensor to assess its detection selectivity and sensitivity. The WHW peptide sequence has specific binding affinity to a conjugated double bond such as aromatic compounds. The binding affinity is also subjected by polarity in many cases, such as, in simple organic solvents [13]. To examine the ability of discriminating different organic solvents, we selected benzene, cyclohexane, cyclopentane, ethylcyclohexane, and isobutyric acid. During all measurements, M-13 bacteriophage-based color sensor has three separate color bands (bands 1–3), which showed color changes that were dependent of solvent type as well as concentration. Fig. 3a shows the results of measuring the least polar solvent, benzene, at different concentrations between 10 and 100 ppm, and was analyzed by color comparison between the initial and final stages. RGB signal changes (ΔRGB) shows the gradual intensification at chromaticness. The intensity of each RGB color component showed the consistent transition with increase of concentration (Supplementary Fig. S1a). The solid line of Fig. 3b corresponds most distinct ΔRGB of each band to the calculated fitting curve obtained using Hill's equation. We obtained a dissociation constant (K_d) of 23.9 ppm for the interaction between benzene and M-13 bacteriophage (Fig. 2b). Fig. 3c shows color changes of a M-13 bacteriophage-based color sensor exposed to 5 different organic solvents. ΔRGB intensity values of each color band are presented in Fig. S1b. All solvents showed unique trends of color change and ΔRGB intensity was related with the structures and polarities of organic solvents. In Fig. 3a and c, each band was calculated and reproduced by the difference of color components from the initial stage to the final stage of each concentration in consistent temperature sectors. To study the discrimination ability of the M-13 sensor, principal component analysis (PCA) was used to analyze the colors as shown in Fig. 3d, showing the two-dimensional scatter plot constructed using the first and second scores. The first two discriminant factors account for 94.17% of the variance in the measurements. As was expected, the score distributions of cyclopentane and cyclohexane were close, but those of isobutyric acid and ethylcyclohexane differed from that of cycloalkane due to the polarity differences. This result confirms that the M-13 bacteriophage-based color sensor is able to discriminate organic solvents.

3.3. Detection of antibiotics

The M-13 bacteriophage-based color sensor was further tested for antibiotics detection, one of the critical environmental pollutants. For this purpose, we selected three most common classes of

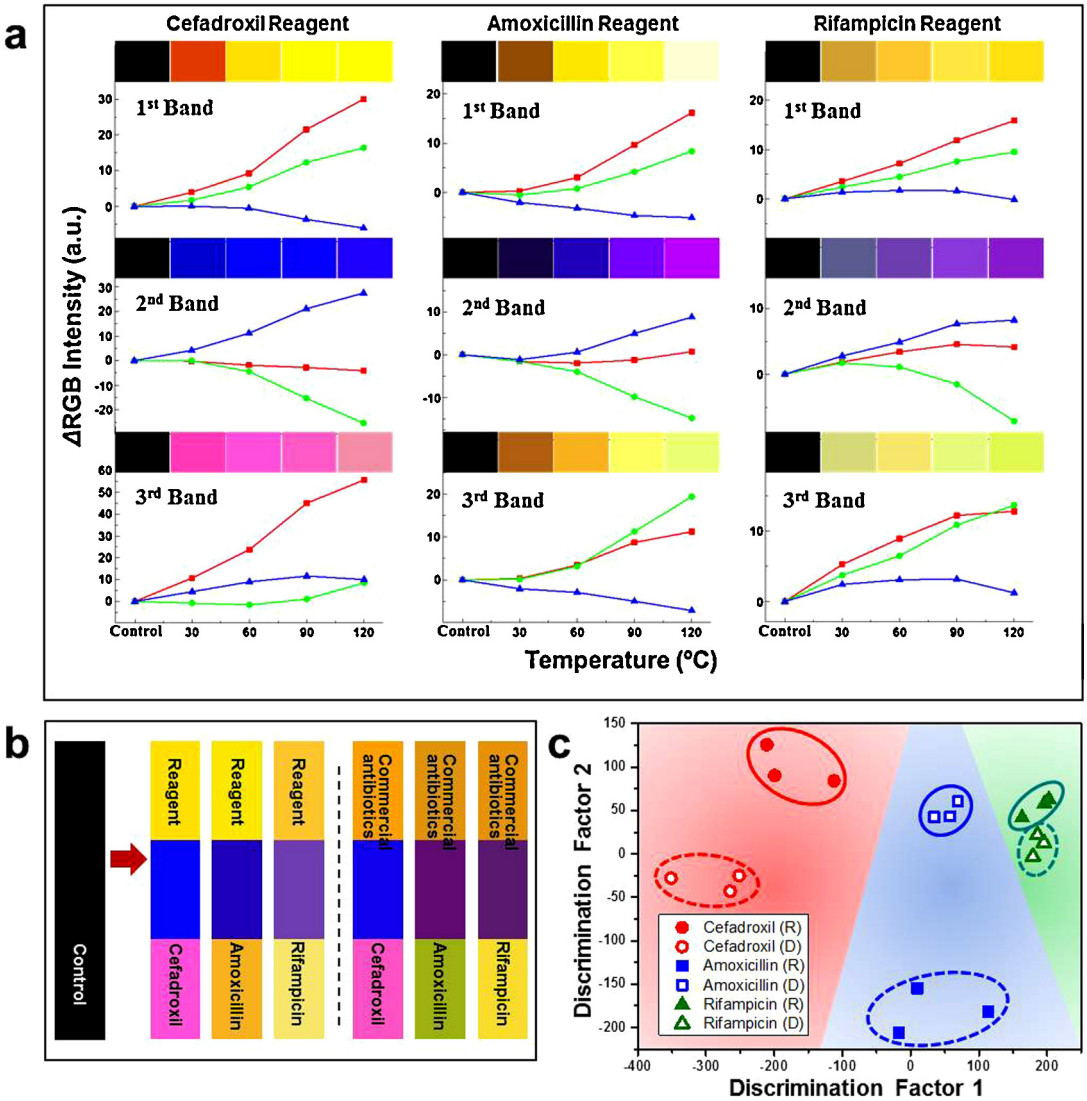

Fig. 4. Measurement of antibiotics using the M-13 bacteriophage-based color sensor. (a) Measurement of antibiotics reagents at different temperatures from 30 to 120 °C. Sample was loaded in the solid phase and heated. The lines guides for the eye. (b) Various antibiotics (reagent and commercial antibiotics) were measured with the color sensor at the same temperature, 60 °C. (c) 2D PCA score scatter plot (the first vs. second principal components) of different antibiotics, where R is reagent and D is commercial antibiotics. Color regions are guides for the eye.

antibiotics, cephalosporin, penicillin and rifampin. In this study, WHW type M-13 bacteriophage was prepared for selective discrimination for these antibiotics since they have different chemical structures as shown in Fig. S2. Powdered antibiotics were spread in the sealed chamber for measurements. Since all experiments were accomplished without direct measuring vapor pressure, we assumed that the sample concentration in gas phase proportionally increased by the temperature elevation. When pure antibiotics were measured with the temperature elevation, the continuous increases in ΔRGB intensity and steady escalations of chromaticness were observed as shown in Fig. 4a. The RGB signal changes by measuring commercial antibiotics using color sensor at different temperatures are shown in Fig. S3. Fig. 4b shows the color change patterns of the tested commercial antibiotics and their pure reagents. All antibiotics in the measurements were clearly differentiated due to the disparity in their chemical structures and functional groups. In addition, the commercial and pure samples exhibited different color patterns. Other components in commercial tablets, such as starch and lactose, influenced these differences. The score scatter plot for the differentiation of two antibiotics is shown in Fig. 4c. Red, blue and green symbols represent cefadroxil, amoxicillin, and rifampicin samples, respectively. A combination of the first and second principal components provided the best discriminatory. The first two discriminant factors account for 86.69% of the variance in the measurements. The positon of cefadroxil and amoxicillin is next each other likely due to the structural similarity, however, still shows decent separation. This PCA result revealed that the detection of antibiotics using the M-13 bacteriophage-based color sensor is selective and sensitive.

4. Conclusion

We successfully demonstrated the M-13 bacteriophage-based color sensor enabling of sensitive and selective detection of antibiotics. The self-assembly property of M-13 bacteriophages allowed us to fabricate a structural color sensor using the simple pulling technique. The pVIII major coat protein of M-13 bacteriophages containing WHW peptide sequence enabled to measure organic solvents as well as antibiotics in ppm level. After focusing on genetic engineering for more stable and functional group friendly M-13 bacteriophage, we could fabricate color sensor applicable to

multiple target chemicals. The PCA results also revealed that the selectivity of M-13 bacteriophage-based color sensor. It is anticipated that the proposed color sensor can be expandable for diverse analytes detection by properly engineered surface on bacteriophage for given applications.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.snb.2016.09.050.

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